Gaza Strip Reconstruction Decision Support System (GSR-DSS)

Project Progress Report - Version 1.0.0

Engineering & Technical Development Team

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1 Executive Summary

The first functional version of the **Gaza Strip Reconstruction Decision Support System (GSR-DSS)** is now complete. This AI-driven platform transforms massive satellite damage data into actionable reconstruction intelligence. By integrating spatial analysis, population dynamics, and infrastructure criticality, the system generates prioritized recovery plans designed to optimize resource allocation in high-impact zones.

2 Key Achievements & Technical Milestones

2.1 Data Processing & Spatial Intelligence

- **Comprehensive Damage Analysis:** Processed 198,308 individual damage points verified via UNOSAT satellite imagery.
- **Hexagonal Grid Mapping:** Segmented the Gaza Strip into 2,159 hexagonal zones (500m radius).
- **Temporal Data Recovery:** Developed "Smart Temporal Fusion" algorithms to recover over 90% of missing values across 14 sensor cycles.

2.2 Infrastructure & Road Network Analysis

- **Segment Mapping:** Analyzed 61,892 road segments using OpenStreetMap (OSM) integrated with damage layers.
- **Critical Connectivity:** Identified 5,598 critical road segments (Damage Level 3-4).
- **Strategic Prioritization:** Applied Edge Betweenness Centrality to identify network bottlenecks.

3 The AI Scoring Framework

The core of the system is the Multi-Criteria Decision Analysis (MCDA) engine, which uses the following Weighted Linear Combination (WLC) formula:

$$Score = \sum_{i=1}^{n}(w_i \times x_i) \cdot \alpha \tag{1}$$

Where:

- w_i: Strategic weight for the criteria.
- x_i: Normalized value of the variable (0-1).
- α: Non-linear activation factor for critical zones.

Table 1: Strategic Weight Distribution

Variable	**Weight (w)**	**Strategic Justification**
Major Hospitals	1.53	Critical for survival and emergency response.
Damage Severity	0.25	Physical requirement for reconstruction.
Temporal Persistence	0.20	Prioritizes long-term suffering areas.
Population Density	0.15	Maximizes the number of beneficiaries.

4 Methodology & Workflow

The system follows a 5-step engineering pipeline:

1. **Data Harmonization:** Merging satellite, OSM, and humanitarian data.
2. **Spatial Aggregation:** Converting points to Hex-Binning managed zones.
3. **Normalization:** Standardizing units into a 0-1 scale.
4. **Network Topology:** Analyzing the strategic importance of road networks.
5. **Unsupervised Clustering:** Using K-Means to categorize 2,159 zones into "Functional Clusters."

5 Next Steps & Road Map

- **[Q2 2026] Web-GIS Interface:** Developing a user-friendly dashboard for stakeholders.
- **[Q3 2026] Live Data Integration:** Connecting to real-time humanitarian feeds (UN OCHA).
- **[Q4 2026] Cost Estimation ML:** Implementing Machine Learning for dynamic price prediction.

6 Conclusion

The **GSR-DSS** is a strategic asset ensuring that every reconstruction effort is backed by scientific evidence, prioritizing the most vulnerable populations and critical infrastructure to build a more resilient future.